

Data Engineering Summer Internship Report.

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Description and Identification

CGI is a leading global consulting and IT services company that provides a comprehensive range of services and solutions to businesses and governments worldwide. The organization's mission is to support its clients in achieving their strategic objectives through innovative technology solutions and consulting expertise. CGI operates within a broad ecosystem, collaborating with other technology firms, software providers, and client organizations to deliver customized solutions that drive operational efficiency and business growth.

During my internship at CGI, I was involved in a project dedicated to creating an automated financial processing system for a client. This initiative was significant as it aimed to revolutionize the client's financial operations by implementing an advanced system that could streamline and automate complex financial workflows. The project sought to enhance accuracy, reduce manual intervention, and improve overall financial reporting efficiency. The financial processing system, being the first of its kind, was designed to address specific client needs while setting a new benchmark for financial automation within the industry.

I felt a sense of responsibility in my role in building and optimizing the data infrastructure necessary for the financial processing system. I worked extensively with Snowflake, a cloud-based data warehousing platform, to design and develop robust data pipelines. These pipelines were essential for extracting, transforming, and loading (ETL) data from various sources, including Salesforce CRM, into Snowflake. I utilized SQL for querying and manipulating data, ensuring that the data processing workflows were both efficient and accurate. Additionally, I employed Python to handle specific data processing tasks and integrate the data seamlessly into the financial system. My contributions to the project included designing the data architecture, implementing ETL processes using Matillion, and developing SQL scripts

to manage and analyze the data effectively. One of the key aspects of my role was to ensure that the data integration from Salesforce CRM into Snowflake was smooth and that the data was accurately staged and processed. This involved writing complex SQL queries to retrieve and manipulate data, as well as using Python scripts to handle more intricate processing requirements.

Throughout the internship, I was able to apply the skills I acquired during my studies, including SQL proficiency, data preparation, and documentation. My academic background in data principles, AI/ML, and business analytics provided a solid foundation for tackling the challenges presented by the project. For instance, my knowledge of SQL and data modeling was crucial for designing efficient data pipelines and ensuring accurate data retrieval. The procedural knowledge I gained from my coursework was instrumental in executing the tasks required for the financial processing system.

As my role evolved, I took on increasing responsibilities, including leading collaborative efforts with team members, and participating in client meetings to refine requirements and address any issues. This experience allowed me to further develop my skills in documentation, code review, and maintaining coding standards. By the end of the internship, I had significantly contributed to the successful development and deployment of the automated financial processing system, aligning with CGI's mission to deliver impactful and innovative solutions to its clients.

Reflection

My experience at CGI was both enriching and illuminating, offering a deep dive into the world of data engineering and financial technology. The opportunity to work on the development of an innovative automated financial processing system was a significant milestone in my

professional journey. The project itself was groundbreaking, not only because it aimed to enhance financial operations for a client but also because it was one of the first of its kind in the industry. Participating in CGI's activities and immersing myself in the company's culture was highly rewarding. CGI is known for its collaborative and dynamic work environment, which was evident in the way team members worked together to achieve project goals. My business unit fosters a culture of innovation and continuous improvement, where ideas are welcomed, and contributions are valued. This supportive environment greatly facilitated my learning and growth throughout the internship.

Engaging with various teams and stakeholders provided me with a broader perspective on how different functions within the organization work together to deliver complex solutions. Regular client and cross-functional meetings were particularly enlightening, as they allowed me to understand client requirements in detail and see how our technical solutions were aligned with their business needs. This aspect of the internship not only improved my technical skills but also enhanced my ability to communicate effectively with clients and team members. The project's focus on building a sophisticated financial processing system allowed me to apply and refine the skills I had acquired during my studies, such as SQL, Python, and data preparation. Working with Snowflake and Matillion in a real-world setting provided me with practical experience in data modeling, ETL processes, and performance tuning. I found it particularly engaging to translate theoretical knowledge into practical solutions, such as designing data pipelines and optimizing SQL queries to ensure efficient data processing.

One of the key lessons I learned was the importance of adaptability and problem-solving in a fast-paced project environment. The challenges encountered during the project, such as integrating data from Salesforce CRM and ensuring data accuracy, required me to think critically and find innovative solutions. This experience reinforced the value of thorough documentation

and the need to maintain high coding standards, which are crucial for successful project execution.

Overall, my time at CGI was a valuable learning experience that broadened my understanding of data engineering and financial technology. The opportunity to work on a pioneering project and contribute to a high-impact solution was both fulfilling and educational. The skills and insights gained from this experience have not only strengthened my technical abilities but also provided me with a clearer perspective on the role of data engineering in driving business success. This internship has significantly shaped my professional development and has inspired me to continue pursuing excellence in the field of data engineering and analytics.

Analysis

During my internship at CGI, several aspects of the project I worked on were notably successful, while others presented challenges that provided valuable learning experiences. Analyzing these elements offers insight into what worked well, the obstacles encountered, and the initiatives that were introduced or created.

Successes and Effective Strategies:

One of the key successes of the project was the effective use of Snowflake and Matillion for building the automated financial processing system. Snowflake's robust data warehousing capabilities allowed for efficient data storage and retrieval, which was crucial for handling the complex financial data involved in the project. Matillion proved to be an invaluable tool for data integration, particularly for bringing data from Salesforce CRM into Snowflake. The seamless integration of these tools ensured that the data pipeline was both efficient and reliable.

Another significant success was the optimization of SQL queries and data processing workflows. By applying the knowledge gained from my studies in SQL and data preparation, I was able to write and refine complex queries that improved the performance and accuracy of data processing. This optimization was crucial for meeting the project's requirement for timely and precise financial data.

Challenges and Obstacles:

Despite these successes, the project faced several challenges. One notable obstacle was the integration of data from Salesforce CRM. Ensuring that the data was accurately transferred and processed in Snowflake required meticulous attention to detail and troubleshooting. The complexity of the data structures and the need for real-time processing presented difficulties that required innovative problem-solving approaches.

Additionally, balancing the various aspects of the project—such as data integration, processing, and performance tuning—proved to be challenging. The dynamic nature of the project environment meant that priorities and requirements could shift, necessitating frequent adjustments to the data pipeline and processing workflows. This required a high level of adaptability and responsiveness.

Initiatives and Improvements:

During the internship, I introduced several initiatives to address these challenges and improve the project's outcomes. For example, I developed a set of standardized templates and documentation practices to streamline the process of data integration and transformation. These templates were designed to improve consistency and reduce the time required for future data processing tasks.

I also took the initiative to implement a more structured approach to performance tuning. By closely monitoring the data workflows and identifying bottlenecks, I was able to make targeted adjustments that enhanced the overall efficiency of the data pipeline. This proactive approach to performance optimization helped to ensure that the system met the required performance benchmarks.

In summary, the internship at CGI provided a comprehensive experience in building and optimizing a financial processing system. The successes achieved, coupled with the challenges encountered, offered valuable lessons in data engineering. The initiatives introduced during the project contributed to its overall success and provided insights into effective strategies for handling complex data processing tasks. The experience underscored the importance of adaptability, problem-solving, and continuous improvement in achieving project goals.

Connection to Coursework

My internship experience at CGI provided a valuable opportunity to apply and expand upon the knowledge I acquired during my studies in Computer Information Systems and Business Analytics. The project involved building an automated financial processing system using Snowflake, SQL, Python, and Matillion, which allowed me to directly apply classroom concepts and skills to real-world scenarios.

Application of Classroom Knowledge:

The foundational skills in SQL and Python that I developed in my coursework were instrumental in my role. For example, the advanced SQL techniques I learned in classes on database management and data manipulation were crucial for optimizing the complex queries required for the financial processing system. Accurately calculating interest rates by using

specific data types, instead of relying on less precise floats, is a direct application of data integrity concepts learned in class. This practical application of SQL not only reinforced my understanding but also highlighted the importance of writing efficient and accurate code for data retrieval and manipulation.

Similarly, the concepts and practices learned in data preparation and cleaning courses were directly relevant to my work with Snowflake and Matillion. My understanding of data staging, transformation, and integration, gained from coursework on data pipelines and ETL processes, was essential for ensuring that the financial data was accurately processed and stored. The ability to critically analyze data quality issues and implement effective cleaning procedures, as taught in my coursework, proved invaluable in maintaining data integrity throughout the project.

The collaborative skills developed in classes on team management and documentation were also beneficial. My role involved leading and participating in cross-functional meetings, creating documentation, and peer reviewing code. The ability to communicate effectively, organize documentation, and work collaboratively were all crucial for the success of the project. This experience reinforced the importance of these soft skills in a professional setting, showing how they contribute to the overall efficiency and effectiveness of project teams.

Identification of Gaps and Areas for Improvement:

While the classroom knowledge provided a strong foundation, the internship also revealed areas where additional learning could be beneficial. One such area is the integration of various data tools and platforms. Although I had theoretical knowledge of data integration, the practical challenges of working with tools like Salesforce CRM and Matillion highlighted the need for a deeper understanding of how different systems interact and the nuances of integrating them effectively.

Another gap identified was in the area of performance tuning and optimization in dynamic project environments. While I had learned the basics of performance tuning, the complexity and scale of the data processing tasks at CGI required a more nuanced approach. This experience highlighted the need for advanced training in performance optimization techniques and strategies, particularly in high-velocity data environments.

Larger Ideas and Field Relevance:

The experience at CGI also connected to larger ideas in the field of data engineering and analytics. The emphasis on building scalable, efficient data systems aligns with broader industry trends toward big data and real-time analytics. The project's focus on automating financial processing reflects a growing demand for automation and efficiency in data management, a trend that is increasingly shaping the future of data engineering.

In conclusion, my internship at CGI allowed me to apply and expand upon the knowledge gained in my academic courses, demonstrating the practical relevance of my studies. It also highlighted areas where further learning could enhance my skills and prepare me for future challenges in the field. The experience underscored the importance of bridging theoretical knowledge with practical application and adapting to the evolving demands of the data engineering landscape.

Conclusion

My internship experience at CGI has been a transformative journey, blending academic knowledge with real-world application and providing a comprehensive understanding of the data engineering field. Throughout my time at CGI, I was deeply involved in building an automated financial processing system using Snowflake, SQL, Python, and Matillion. This hands-on project

not only showcased the practical utility of my classroom learning but also illuminated the complexities and nuances of applying theoretical concepts in a professional environment.

The experience was highly reflective of my academic background in Computer Information Systems and Business Analytics. The skills I acquired in SQL, data preparation, and collaborative work were directly applicable and critical to the success of the project. My role in optimizing data pipelines and integrating systems underscored the relevance of my coursework and reinforced the value of these skills in addressing real-world challenges.

However, the internship also revealed areas for further growth and learning. The practical challenges of integrating diverse data tools and optimizing performance in a dynamic setting highlighted the need for deeper knowledge and advanced techniques. These insights point to the evolving nature of the field and the importance of continuous learning and adaptation.

Overall, the internship at CGI has been a valuable bridge between academic theory and practical application. It has provided me with a clearer perspective on the data engineering field and the skills necessary to excel within it. The experience has not only affirmed my academic learning but also emphasized the need for ongoing development in areas such as tool integration and performance optimization. This reflection on my internship highlights the importance of applying classroom knowledge in real-world contexts and adapting to the ever-changing landscape of data engineering.

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